

# Jianxiang Dong

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## Research Summary

Machine learning engineer and AI researcher with a Ph.D. in Computer Science. Academic research focuses on video-language models, video understanding, temporal grounding and retrieval, multimodal learning, and biomedical image analysis. Industry work focuses on AI agents, LLM/MLLM post-training, training-data curriculum design, multimodal video understanding, and generative AI for video.

## Education

### Ph.D. in Computer Science

*Stony Brook University*

*Advisor: Prof. Zhaozheng Yin*

May 2026

*Stony Brook, NY*

### M.S. in Computer Science

*George Washington University*

May 2020

*Washington, DC*

### B.S. in Automation

*Northeastern University*

Jul 2018

*Shenyang, China*

## Professional & Research Experience

### Machine Learning Engineer

*TikTok (ByteDance)*

May 2026 – Present

*San Jose, CA*

- **Short-Drama Script AI – AI Agents & LLM Post-Training:** Built an AI agent and evaluation harness for long-form short-drama script quality assessment using structured rubrics, task-specific context and memory, and LLM-as-a-judge workflows. Also contributed to an end-to-end generation pipeline combining LLM post-training with agentic workflows for creative concept and full-script generation.
- **Short-Drama Remake – Video Understanding & Generative AI:** Developed a multimodal video-understanding and generative-AI pipeline for cross-market short-drama localization, combining face detection, recognition, tracking and identity aggregation with visual prompting, VLM-based asset extraction, storyboard reconstruction, and video generation.
- **AI Canvas – Data Curriculum & LLM/MLLM Post-Training:** Built data curriculum and post-training pipelines for text/image-to-HTML visual generation, including behavior-aware data curation, synthetic-data generation, multi-model judging and targeted arbitration, supervised fine-tuning, visual-quality evaluation, quantization, and deployment.

### Research Assistant

*Computer Vision Lab, Stony Brook University*

Jan 2022 – May 2026

*Stony Brook, NY*

- **Video-Language Modeling and Video Understanding:** Developed deep-learning models for temporal sentence grounding, text-to-video retrieval, and grounded video question answering, using graph-based, contrastive, boundary-aware, and multimodal-LLM approaches to improve retrieval and localization performance over prior baselines.
- **Annotation-Efficient Learning:** Proposed weakly and semi-supervised learning frameworks with active-learning strategies, achieving performance comparable to fully supervised systems using roughly 30% of full temporal annotations; published in IEEE TCSVT and IEEE TMM.
- **Biomedical Image Analysis – Brookhaven National Laboratory Collaboration:** Conducted Stony Brook RA research on a collaborative BNL project in macromolecular crystallography, developing deep-learning methods for X-ray diffraction spot segmentation and content-aware image compression; improved spot recall by approximately 12% and achieved approximately 50× compression with negligible loss in downstream detection accuracy.

### Research Intern

*Xeo Air*

Jul 2022 – Dec 2022

*St. Louis, MO*

- Developed computer-vision systems for multi-class bridge-defect detection and segmentation, together with an expert-in-the-loop workflow for iterative model improvement.

## Peer-Reviewed Publications

- **Annotation-Efficient Hybrid Learning for Temporal Sentence Grounding**  
Jianxiang Dong, Zhaozheng Yin  
*IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 36(2):2594–2606, 2026

- **Weakly Semi-Supervised Temporal Sentence Grounding in Videos with Point Annotations**  
Jianxiang Dong, Zhaozheng Yin  
*IEEE Transactions on Multimedia (TMM)*, 28:2268–2278, 2026
- **Content-Aware Image Compression Model for Macromolecular Crystallography Data**  
Jianxiang Dong, Zhaozheng Yin, Herbert J. Bernstein, Jean Jakoncic  
*IEEE International Symposium on Biomedical Imaging (ISBI)*, Oral Presentation, 2026
- **Graph-Based Dense Event Grounding with Relative Positional Encoding**  
Jianxiang Dong, Zhaozheng Yin  
*Computer Vision and Image Understanding (CVIU)*, 251:104257, 2025
- **Bragg Spot Finder (BSF): Machine-Learning-Aided Diffraction Spot Filtering**  
Jianxiang Dong, Zhaozheng Yin, Dale Kreidler, Herbert J. Bernstein, Jean Jakoncic  
*Journal of Applied Crystallography*, 2024
- **Boundary-Aware Temporal Sentence Grounding with Adaptive Proposal Refinement**  
Jianxiang Dong, Zhaozheng Yin  
*Asian Conference on Computer Vision (ACCV)*, 2022

## Manuscripts Under Review

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- **RoCon: Relieving Over-Concentration Problems for Partially Relevant Video Retrieval.** Jianxiang Dong, Zhaozheng Yin. Under review, CVPR 2027.
- **Multi-Scale Token-Aware Contrastive Learning with Hard Negative Mining for Long-Video Temporal Sentence Grounding.** Jianxiang Dong, Zhaozheng Yin. Under review, IEEE Transactions on Multimedia (TMM).

## Academic Service & Teaching

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- **Peer Reviewer:** AAAI.
- **Teaching Assistant:** Introduction to Java Programming; Introduction to Computer Networks, Stony Brook University.
- **Research Mentor:** CSIRE deep-learning program, Stony Brook University.